

1) a) Check if $G(Fp) \equiv F(Gp)$.

$G(Fp)$ means that p is true infinitely often. In other words, for any state, there exists a future state where p will be true. On the other hand, $F(Gp)$ means that at one point, p will be eternally true.

These two statements are not equivalent. The counter example for this is:

$$\varepsilon (p \sim p)^\omega$$

This implies, on infinite sequence: $p, \sim p, p, \sim p, \dots$

This sequence satisfies $G(Fp)$ but does not satisfy $F(Gp)$. Hence:

$$G(Fp) \not\equiv F(Gp)$$

b) Check if $p \vee q \equiv (p \vee q) \wedge G((p \wedge \sim q) \Rightarrow X(p \vee q))$

$p \vee q$ means that q should eventually happen, and p must hold until then. $(p \vee q) \wedge G((p \wedge \sim q) \Rightarrow X(p \vee q))$ means that we must start with p or q being true. Then, for all other subsequent states, whenever we have p but not q , the next state must have either p or q .

These two statements are not equivalent. The counter example for this is:

$$\varepsilon (p)^\omega$$

This implies, on infinite sequence: $p, p, p, \dots$

This sequence does not satisfy $p \vee q$ but satisfies

$(p \vee q) \wedge G((p \wedge \neg q) \Rightarrow X(p \vee q))$. Hence:

$$p \vee q \neq (p \vee q) \wedge G((p \wedge \neg q) \Rightarrow X(p \vee q))$$

2) The Buchi Automaton is formally defined as:

$$A = (Q, S, \Delta, F)$$

where:

- Q is a finite set of states
- $S \subseteq Q$ is the set of initial states
- $\Delta \subseteq Q \times A \times Q$ is the set of transitions
- $F \subseteq Q$ is the set of final states

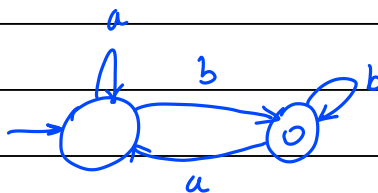
The language $L(A)$ of a Buchi automaton consists of all infinite words that visit a final state infinitely often.

Hence, the complement, $L(\bar{A})$ must consist of words that visit F only finitely often. Therefore, to construct a Buchi automaton for $L(\bar{A})$, we must:

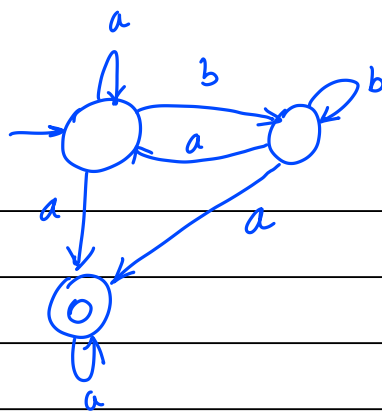
- i) Visit the final states for a finite number of times.
- ii) Transition to a state that will never visit the final state

An example:

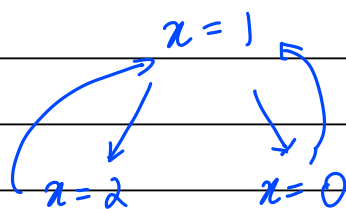
Buchi automaton for
infinitely many b 's
(from slides)



The complement automaton can be as follows:



3)a) We can describe this system using a flowchart:

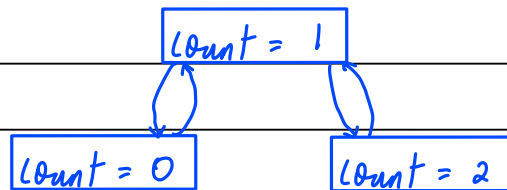


(initial state)

(both guards are true, any one will be chosen)

Hence, the system can be described by this sequence, 1, 0 or 2, 1, 0 or 2, ...

This can also be written as an automaton:



b) No. The model is satisfied by $(10)^w$, which represents the sequence $\text{count}=1, \text{count}=0, \text{count}=1, \text{count}=0, \dots$. $\text{count}=2$ never occurs in this sequence. Hence, $x=1 \not\models F(x=2)$.

c) The formula $x=1 \Rightarrow F(x=2)$ can be written (using p & q) as:

$$p \Rightarrow Fq$$

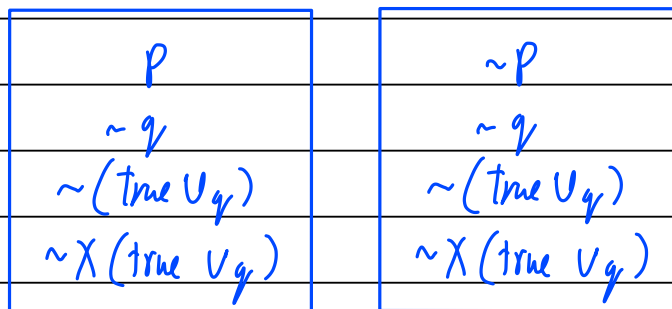
Let us obtain the negation of this formula:

$$\begin{aligned}
 \sim(p \Rightarrow F_q) &\equiv \sim(\sim p \vee F_q) \quad [a \rightarrow b \equiv \sim a \vee b] \\
 &\equiv \sim(\sim p) \wedge \sim(F_q) \quad [\sim(a \vee b) \equiv \sim a \wedge \sim b] \\
 &\equiv p \wedge \sim(\text{true} \vee q)
 \end{aligned}$$

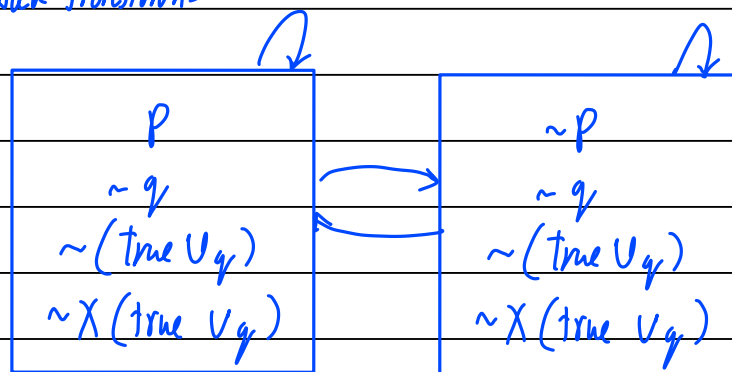
Let us now construct the Buchi automaton for $\varphi = p \wedge \neg q$

i) Obtain closure: $\mathcal{d}(\varphi) = \{p, q, \text{true} \vee q, \sim p, \sim q, \sim(\text{true} \vee q), X(\text{true} \vee q), \sim X(\text{true} \vee q)\}$

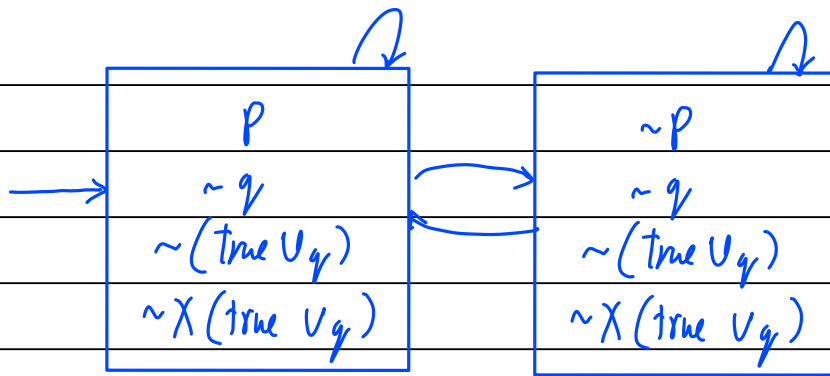
ii) Form atoms (states):



iii) Add transitions:

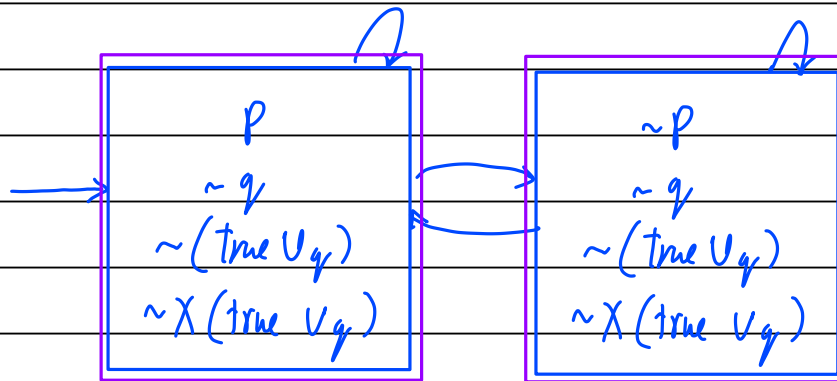


iv) Label initial states:

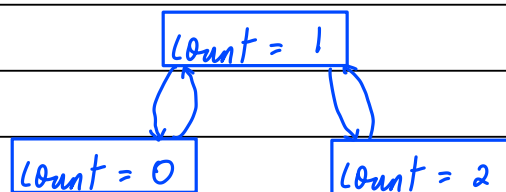


v) Add accepting states (both states are accepting)

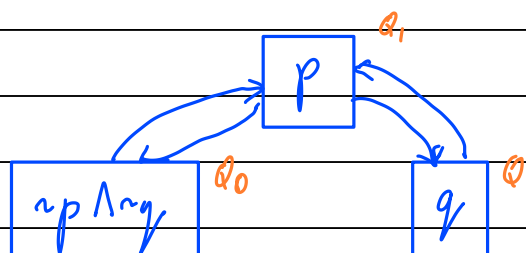
$$F = \{ A \mid (\text{true} \vee q) \wedge A \quad \forall q \in A \}$$



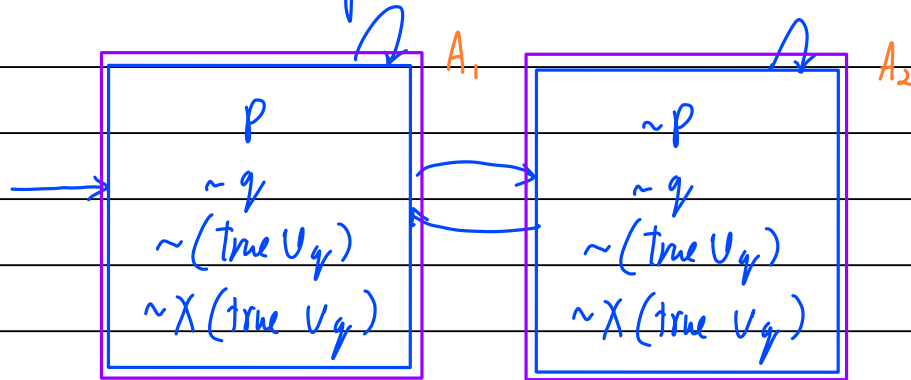
d) The transition system is as follows:



We can re-write this in terms of p & q (for compatibility with the Buchi auto mth):



The Buchi automaton is as follows:



Let us identify the products (so that we can identify which cells are possible):

$Q_0 \times A_1 : X$

$Q_0 \times A_2 : \checkmark$

$Q_1 \times A_1 : \checkmark$

$\sim p$
 $\sim q$
 p
 $\sim q$
 $\sim(\text{true} \cup q)$
 $\sim X(\text{true} \cup q)$

$\sim p$
 $\sim q$
 $\sim p$
 $\sim q$
 $\sim(\text{true} \cup q)$
 $\sim X(\text{true} \cup q)$

p
 p
 $\sim q$
 $\sim(\text{true} \cup q)$
 $\sim X(\text{true} \cup q)$

$Q_1 \times A_2 : X$

$Q_2 \times A_1 : X$

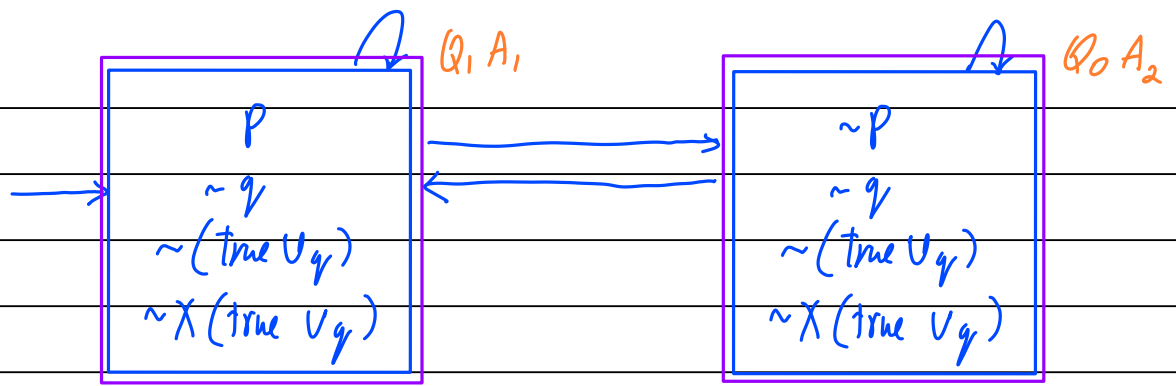
$Q_2 \times A_2 : X$

p
 $\sim p$
 $\sim q$
 $\sim(\text{true} \cup q)$
 $\sim X(\text{true} \cup q)$

q
 p
 $\sim q$
 $\sim(\text{true} \cup q)$
 $\sim X(\text{true} \cup q)$

q
 $\sim p$
 $\sim q$
 $\sim(\text{true} \cup q)$
 $\sim X(\text{true} \cup q)$

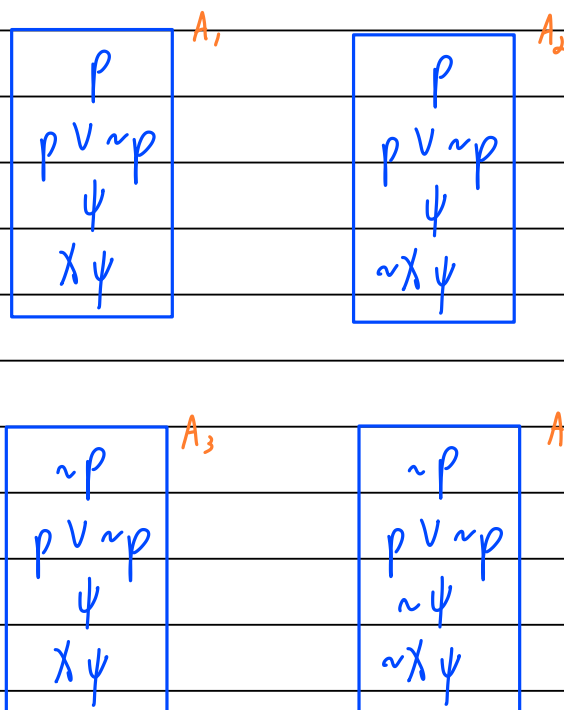
Hence, the only possible states are $Q_0 \times A_2$ and $Q_1 \times A_1$. Let us construct the automaton:



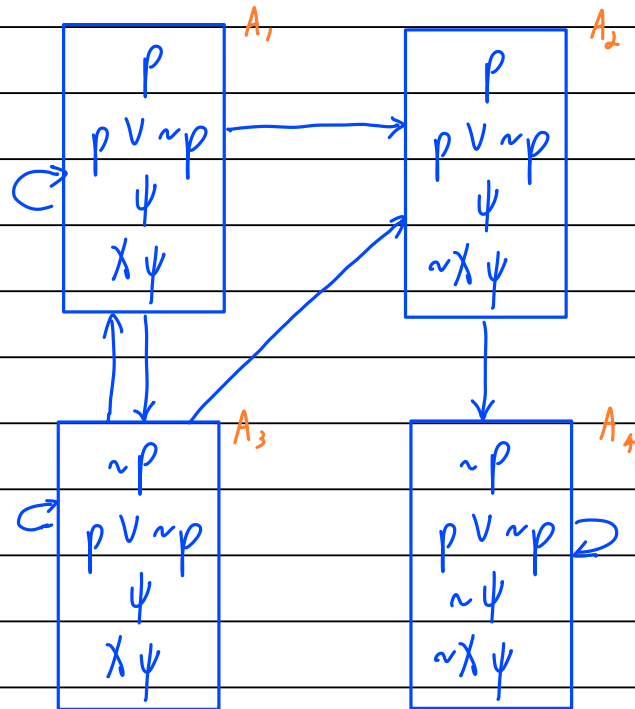
4) Let $\varphi = \text{true} \cup p \equiv (p \vee \sim p) \cup p$

i) $\mathcal{Q}(\varphi) = \{p, \sim p, p \vee \sim p, \sim(p \vee \sim p), \varphi, \sim \varphi, X\varphi, \sim X\varphi\}$

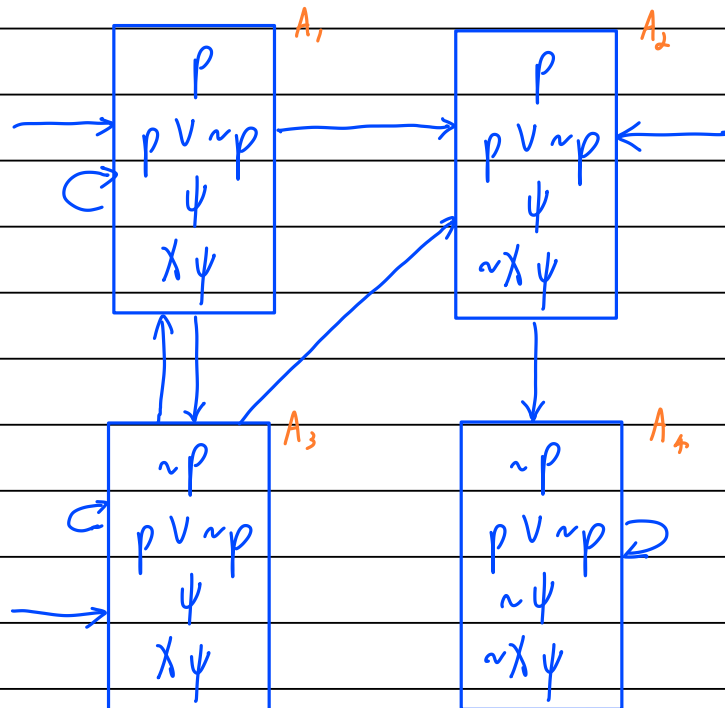
ii) Form Atoms:



iii) Add transitions:



iv) Add initial states: $A_1, A_2 \in A_3$ are all initial because they contain ψ .



v) Add final states: $A_i \in F$ iff $\psi \in A_i \vee p \in A$

